HEALTH-A-THON 2026 · CANCER TRACK · CLINIC OPERATIONS & PATIENT FLOW

RAD-SMART

The right patient, on the right machine, at a predictable time.

Radiotherapy AI-assisted Dynamic Scheduling, Machine Allocation and Resource Tracking



One planned treatment day: each block is a patient's session (routine, new start, complex, blood irradiation)

Dr Akshay Dinesan, Manipal: team leader and doctor partner · **Abhinand T M**: technical lead · **P S Raghotham Rao**: lead engineer

Dr Shirley Lewis Salins: doctor partner · **Dr Umesh Velu**: doctor partner

Assistive, not diagnostic

THE PROBLEM

Every weekday, 70–90 patients wait for one machine

Appointment times are given by hand, without knowing how many machine-minutes each patient needs. Too many people are told to come at once, and they wait, every day of a 5–7 week course, usually with a caregiver.

OUR DEPARTMENT'S RECORDS · 55 DAYS, OCT–DEC 2024 · 3,137 PATIENT-DAYS

109 min

one patient in ten waits longer than this, from arrival to treatment

22%

treated more than an hour after their appointment; only 29% within 15 minutes

19 → 66 min

median wait, arriving before 10:00 against after 19:00: delays build all day

The load is rising, from 54 patients a weekday in October to 67 in December, and busy days are worse. 69% of India's 554 radiotherapy facilities have only one machine.

Sources: department records (anonymised, aggregates only); problem statement (Dr A. Dinesan); J Med Phys 2024

WHY IT IS HARD

Allocation under hard rules and soft priorities

Hard rules: never broken

- New starts finish by 17:00
- SBRT, SRT, TBI, CSI in 10:00–17:00
- Blood irradiation 13:30–14:00, daily
- 1 breast board, 1 ABC, shared with CT
- MHRC: 16:00–17:00, then the 17:00 bus
- Public transport home by 21:00
- Urgent starts treated by 18:00
- One standing slot for the whole course

Soft priorities: traded off

- Paying patients' preferred times
- Older patients earlier
- Ward, dormitory, nearby: late evening
- Complex cases in 12:00–13:30
- A day of 08:30 to 01:00, little overtime

And every day brings events: breakdowns of 15 minutes to 48 hours, overruns, no-shows, urgent add-ons and imaging holds, each of which should trigger a re-plan.

FIVE APPROACHES

We compared five ways to build it

Approach	Strength	Weakness	Score
Rule-based slot templates	Simple, transparent	Can't absorb variation or faults	3.10
LLM-only agent	Conversational	No guarantee on hard rules	2.60
Static optimisation	Guarantees hard rules	No learning or live recovery	3.60
Deep reinforcement learning	Adaptive in principle	Opaque; needs huge data	2.45
Hybrid (selected): predict, optimise, simulate, explain	Best of each technique	More parts, built in layers	4.85

Weighted score out of 5: safety and scope 20%, impact 20%, real-time 15%, feasibility 15%, explainability 10%, ease and languages 10%, configurability 10%.

THE SOLUTION

How RAD-SMART works, and who decides

1 · PREDICT

Minutes, not head-counts

Expected and cautious (P80) minutes per session, starting from the department's estimate table.

PoC: 2.1 min error vs 2.6 for averages

2 · OPTIMISE

Hard rules guaranteed

A solver gives each course a standing slot and repairs the day live; hard rules cannot break.

PoC: 87 patients, all 8 rules re-checked

3 · SIMULATE

Tested before it is used

A twin calibrated on the department's records tests every plan and forecasts three weeks of load.

PoC: 300 days; 2 machines, peak load 154% → 98%

4 · EXPLAIN

Kannada, Tulu and more

A Sarvam AI copilot explains placements, edits rules from plain language and messages patients.

The LLM never schedules

PEOPLE DECIDE

The doctor approves each course and its standing slot · the technologist approves the day and records the times · a breakdown or urgent start triggers a re-plan in seconds, approved by a technologist or doctor before anyone is messaged · actual times retrain the model weekly

Runs inside CARE or the hospital EMR on a hospital server · Excel list, register photo (OCR), QR check-in, WhatsApp

BUILT INTO CARE

From the visit page to the treatment day

DOCTOR

Clicks **RADSMART** on the visit page, enters the prescription and ticks a box for every rule. RAD-SMART proposes a new start before 17:00 and one standing slot for the whole course; the doctor approves.

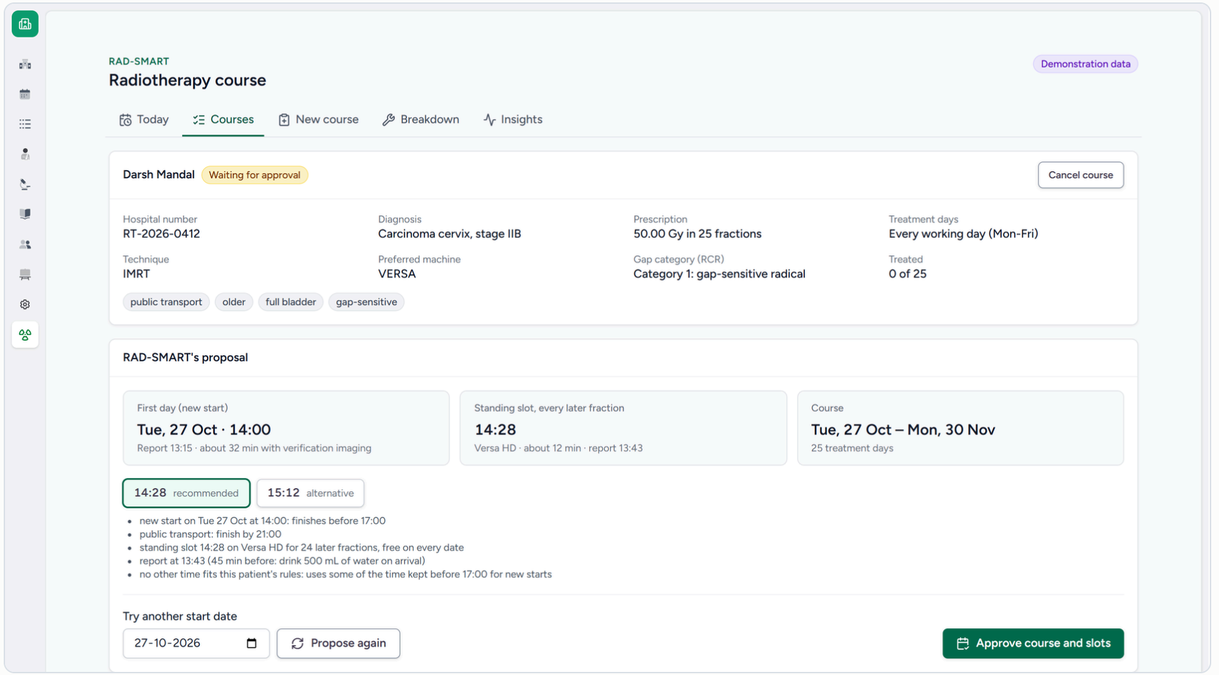
TECHNOLOGIST

Runs the day: rule check, timeline, **Arrived → In room → Out**, machine status and slot-change requests.

BREAKDOWN, 15 MIN TO 48 H

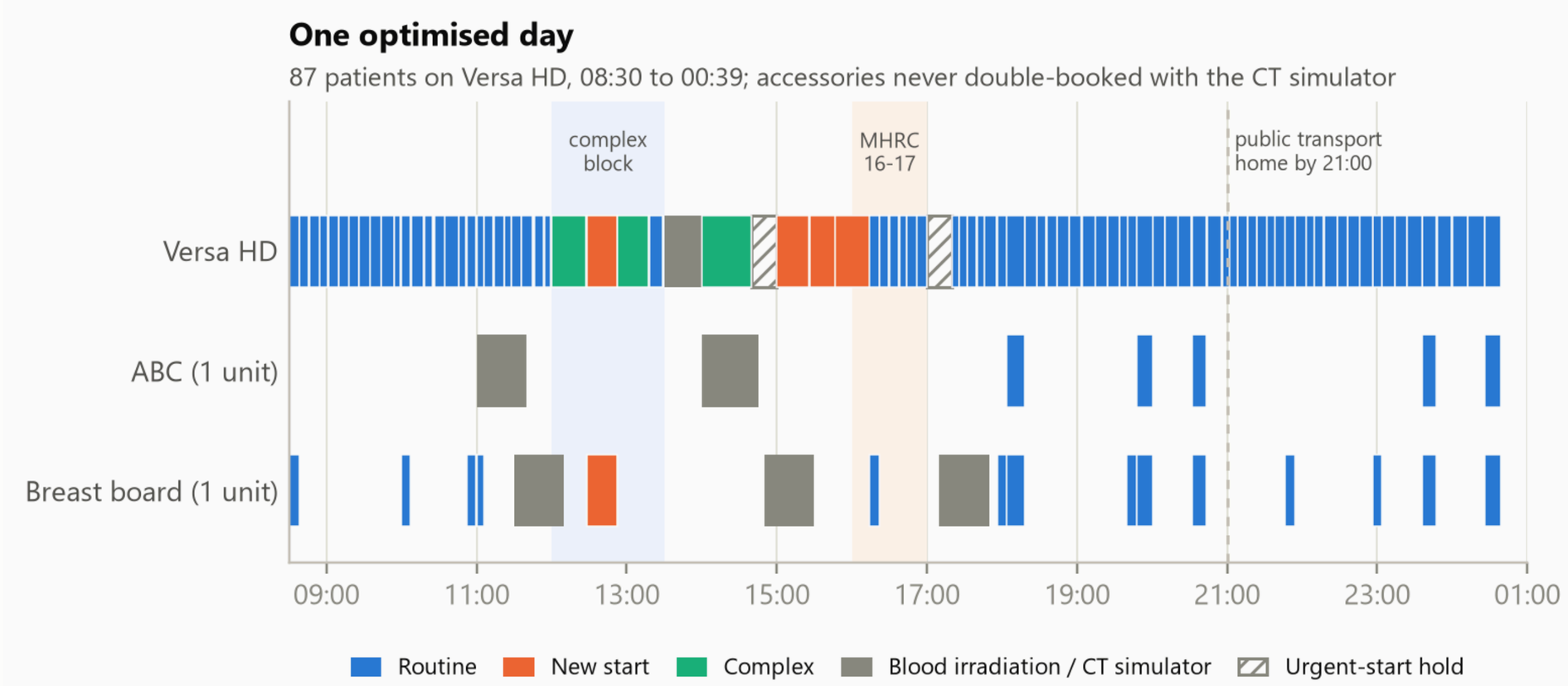
Up to 60 min: everyone treated, with overtime. Longer: only as many sent back as needed; patients at home told to wait. Approved before anyone is told.





PROOF OF CONCEPT

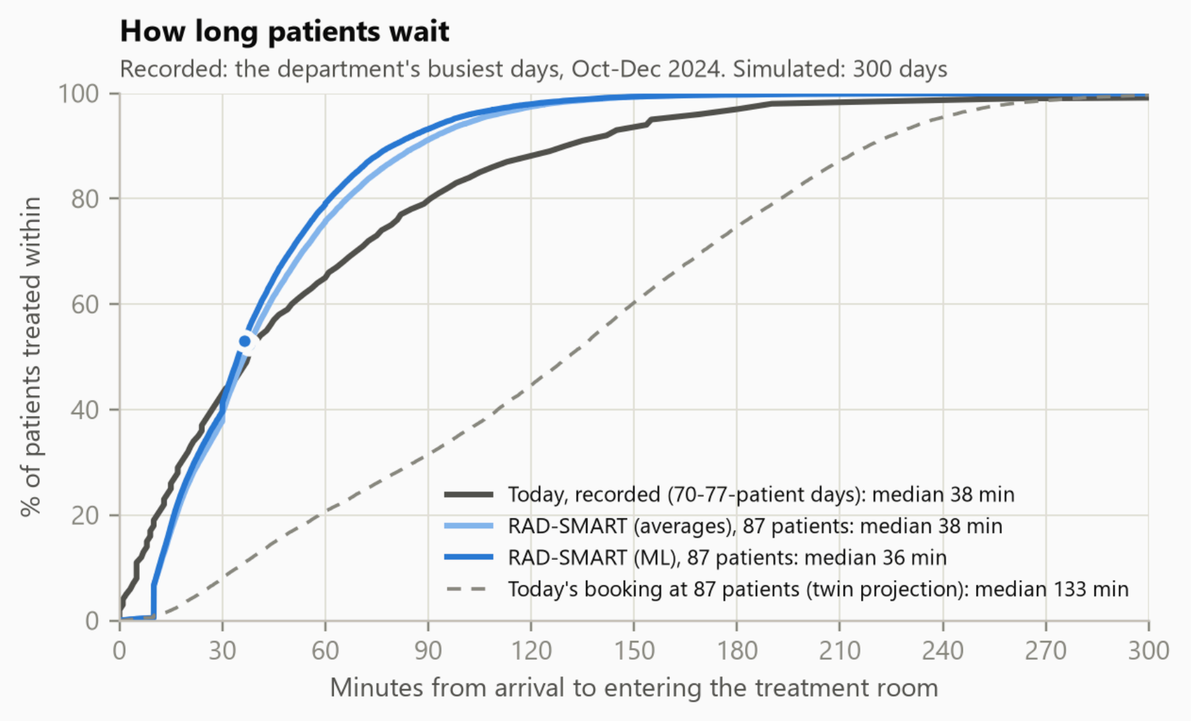
One optimised day, with every rule held



Synthetic day: 87 patients, 890 of 960 machine-minutes (93% load); all 8 hard rules met, none unplanned

PROOF OF CONCEPT · 300 SIMULATED DAYS AGAINST THE RECORDS

Fewer long waits, and times that hold



130 → 76 min

90th-percentile wait, against the busiest recorded days

27% → 6%

treated more than an hour after their time

66 → 36 min

median wait for evening arrivals: no build-up

104 → 39 min

150-min machine fault: the same plan without → with live re-planning

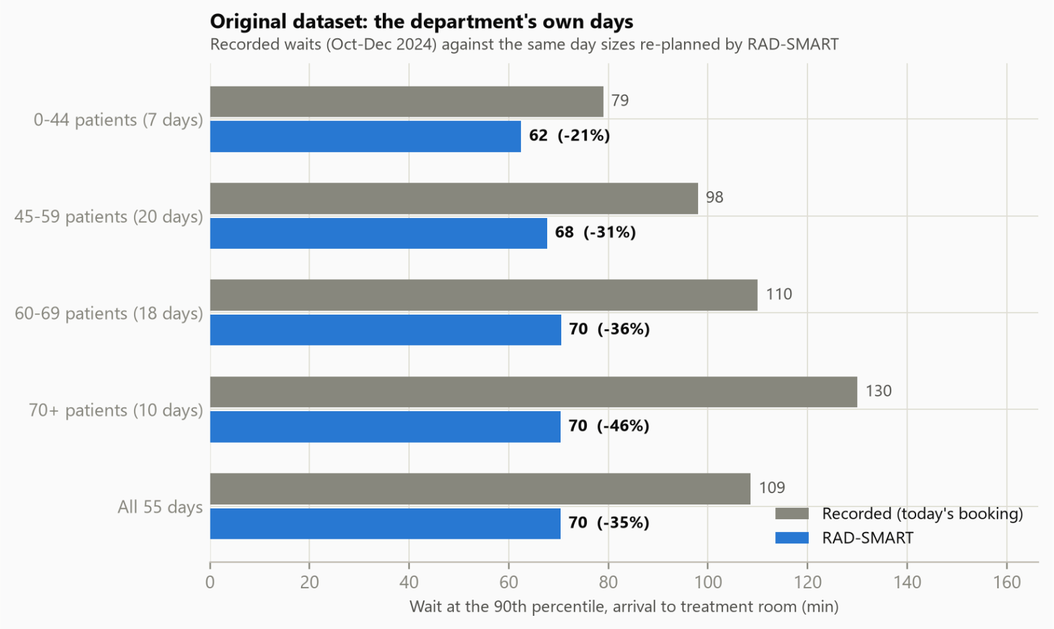
The typical wait stays about the same (38 → 36 min, preparation included), and 53% are treated within ±15 min of their time instead of 30%. What goes is the long, unpredictable wait.

Today: department records, 70–77-patient days. RAD-SMART: synthetic 87-patient day in a twin calibrated on the records

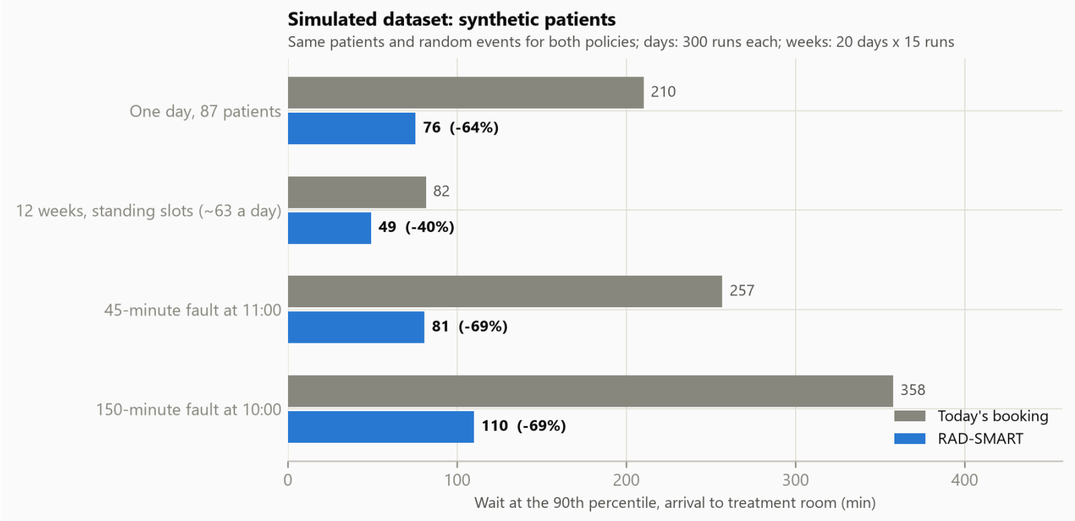
WITH AND WITHOUT RAD-SMART

Both datasets: far fewer long waits, at a price

Original: the department's 55 days, 108.6 → **70.4 min (–35%)**



Simulated: 12 weeks of standing slots, 82 → **49 min (–40%)**



The price, over 12 weeks: the last patient finishes about 00:07 instead of 21:55 · one new start in five begins a little later (0.4 days) · a time the patient asks for is kept less often · 8 min overtime a day

GUARDRAILS AND PILOT

Safe by design, then 90 days on the Versa HD

Safe by design

- **Proposal until approved.** The doctor approves every course; a technologist or doctor every breakdown response
- **Rules re-checked.** An independent checker verifies all 8 hard rules, and each standing slot, before publishing
- **Clinicians decide.** No diagnoses, doses, images or notes are read
- **No identifiers to AI.** Models see pseudonymous IDs; names stay on the hospital server
- **Full audit trail.** Every plan, override, rule version and message is logged
- **Synthetic and anonymised only.** Records are used as aggregates; DPDP-aligned

Days 1–14
Baseline
Room entry and exit taps added

Days 15–35
Shadow mode
RAD-SMART plans nightly; RTTs compare, nothing changes

Days 36–90
Live
Doctors approve courses, technologists run the day in CARE; patients get times and live updates

PRIMARY KPI

≥ 25% lower

90th-percentile wait from arrival to linac entry (records: 109 min)

ALSO TRACKED

- Over an hour late: 22% → 10% or fewer
- At least 45% within ±15 min (today 29%)
- Median wait, overtime and throughput no worse
- Hard rules kept; no group worse off

Now: the app runs in CARE · **next:** patient messaging, then a pilot in Q1 2027 · **our ask:** a pilot slot with mentors, Sarvam API credits, NCG introductions

Interrupted time-series analysis · about ₹2–3 lakh: server, 3 tablets, QR printer, messaging